

Sunny Patel

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TECHNICAL SKILLS

Programming: Java, Python, Go, Scala, PowerShell, C++, C#, YAML, Kotlin, Assembly (x86/ARM)

Backend: Django, Ruby on Rails, MongoDB, PostgreSQL, MySQL, Express.js, ASP.NET Core, Spring Boot, Kafka

Frontend: React.js, JavaScript, Flutter, TypeScript, WebGL, GraphQL, Tailwind CSS, Three.js, Vue.js

DevOps/Cloud: Git, Docker, Kubernetes, Azure, GCP, AWS (ECS, S3, Lambda, Beanstalk, RDS), Jamf Pro, Datadog

OS: Windows Server, Kali Linux, Debian, Red Hat Enterprise Linux, FreeBSD

EDUCATION

Ontario Tech University

Honours Bachelor of Science in Computer Science

Oshawa, ON

2022 – Present

EXPERIENCE

Software Developer – IBM

Jan. 2026 – Present

Markham, ON

Hybrid

- Built and shipped features across the **full stack** of an AI-powered sales enablement platform, spanning backend APIs, task orchestration, database migrations, frontend, and cloud infrastructure.
- Integrated **watsonx** foundation models into platform workflows, including **RAG pipelines**, real-time LLM-driven internationalization, **vector-based semantic search**, and prompt engineering for AI-assisted content tooling.
- Led **multi-repository integrations** across internal platform services, designing shared API contracts and decoupled dependencies built for maintainable service-to-service evolution, consumed by engineering teams across the organization.
- Deployed to IBM Cloud with full operational visibility, shipping feature releases, performance fixes, **outage resolution**, and **data migrations** across staging and production environments.
- Collaborated with non-technical stakeholders to scope, build, and iterate on **client-facing** tailored AI tools and internal workflows, translating business requirements into delivered systems with continuous revision cycles.

IT Technician – Canada's Wonderland

Jun. 2023 – Jan. 2026

Vaughan, ON

On-site

- Supported pre-cutover **tenant-to-tenant migration** during Six Flags acquisition for 3,000+ directory objects, including **hybrid Azure AD/on-prem onboarding**, OU staging, and **Ivanti Neurons UEM** rollout.
- Authored 10+ **PowerShell/ConnectWise** scripts automating workstation imaging, printer/DFS drive mappings, and user profile migrations.
- Built and deployed **MDT** task sequences and Sysprep-captured **WIMs** for Windows fleet provisioning.
- Managed **Active Directory** accounts, security groups, and **GPOs**, ensuring consistent ACL enforcement across **Exchange**, **SharePoint**, and internal apps.
- Configured **CUCM**, **Unity**, and **Finesse** to optimize VoIP routing, voicemail for call center operations.
- Performed network maintenance including switch patching, **VLAN tagging**, trunking, and **PoE resets** via **SSH** in PuTTY.
- Deployed and supported **Oracle Symphony/EMC POS**, debit, and KDS systems, with rollouts tracked in **Jira/Confluence**.

System Support Specialist – Mackenzie Health

Jan. 2023 – Aug. 2023

Richmond Hill, ON

On-site

- Migrated 400+ **Surface tablets** to bedside **iPads**, leveraging **Intune** and **Jamf Pro** for secure deployment across clinical units.
- Developed internal **admin portal** for managing app services, user roles, and subscription lifecycles.
- Implemented **Vocera VOIP** on iPhones to enable real-time communication across ER and nursing staff.
- Coordinated with vendors on bug logs, updates, and hardware issues, ensuring minimal downtime for critical hospital devices.

PROJECTS

- basalt – GPU Hazard Verifier** | *Python, CUDA, GPU Assembly (SASS), ELF/cubin*   2026 – Present
- Built the **world's first** tool that can prove GPU machine code it did not compile is safe to run, a check **NVIDIA itself does not ship**: nothing on Blackwell (**RTX 50 series**) validates its 21 scheduling bits, so one cycle short silently returns a **wrong answer**.
 - Audited **2,762 kernels NVIDIA bundles inside CUDA**, code it had never seen: **0 errors across 10.2M dependencies**, and **0 missed** on 233 kernels broken on purpose.
 - Proved it against the vendor's bytes: an assembler **wrong 0 times across 5.2M** vendor instructions, and a from-scratch scheduler **byte-identical on 439 of 439** GPU kernels.
 - Reverse-engineered an **undocumented instruction set** with no vendor spec, deriving **345 forms across 90 opcodes** by flipping single bits and measuring latency on silicon by **fault injection**; published with a **Zenodo DOI** and open-sourced on **PyPI**.
- Sunnify Spotify Downloader** | *PyQt, yt-dlp, FFmpeg, GitHub Actions, Sigstore, Next.js*   2024 – Present
- Creator and lead maintainer of a cross-platform Windows/macOS/Linux app with **40,000+ downloads** and **700+ stars**; reverse-engineered Spotify's web player to pull full **1000+ song** playlists with **no login, account, or API key** by parsing its hidden `__NEXT_DATA__` payload and reusing its anonymous access tokens.
 - Built the **PyQt** client and **headless CLI** over a shared engine, downloading with **4 workers in parallel** while staying responsive through thread-safe UI updates and disciplined background-thread management under load.
 - Engineered the audio engine on **yt-dlp** and **FFmpeg** across **5 formats**, with titles, artwork, and tags (**ID3**, MP4, Vorbis) formatted to display correctly on phones, cars, and desktop players, plus logging that pinpoints why any download fails.
 - Secured the release pipeline to **SLSA Build Level 3**, a supply-chain security standard most production companies never reach, so every download ships with a tamper-proof, cryptographically signed proof of authenticity (**Sigstore**, software bill of materials, automated vulnerability scanning).
 - Automated the entire build-and-release flow with **GitHub Actions**, publishing to **Homebrew** and one-click Windows/Linux installers alongside a companion **Next.js** site, with every release shaped by user-filed issues rather than a roadmap.
- ATS Screener** | *SvelteKit, Svelte 5, TypeScript, LLM, Astro, Firebase*   2026 – Present
- Launched a free, open-source alternative to paywalled ATS checkers; reached **1,500+ users within 24 hours** with **zero marketing spend**, architected to run at **\$0 infrastructure cost**.
 - Modelled **6 enterprise ATS platforms** (Workday, Oracle Taleo, iCIMS, Greenhouse, Lever, SAP SuccessFactors) as distinct simulators rather than one generic checker, each with its own parser strictness, keyword strategy (exact/fuzzy/semantic), **6 weighted scoring dimensions**, and executable quirk rules reproducing real vendor parsing behaviour.
 - Wrote the resume-to-job matching engine from scratch instead of calling an embeddings API: a custom tokenizer, **TF-IDF** term weighting, and a hand-curated skills taxonomy that collapses variants into canonical terms, keeping matching **deterministic, explainable, and free to run**.
 - Engineered a dual-mode scoring engine backed by **470+ unit tests**: **LLM-based analysis** as primary with a fully deterministic rule-based fallback, plus a **cross-vendor provider chain** across independent quotas, per-IP rate limiting, and content-addressed response caching.
 - Implemented **privacy-first client-side parsing** via pdfjs-dist (PDF) and mammoth.js (DOCX); resume files never leave the browser, only extracted text reaches analysis.

CERTIFICATIONS

- IBM RAG and Agentic AI Specialization** Issued: Feb 2026
- Skills: RAG Architectures, Vector Databases, LangChain, Agent Design, Tool Calling, Prompt Engineering
- Microsoft Full-Stack Developer Specialization** Issued: Feb 2026
- Skills: Full-Stack Architecture, ASP.NET Core, React, REST APIs, Authentication, Cloud Deployment
- Microsoft GH-300: Github Copilot Intermediate** Issued: Oct 2025 – Expires: Oct 2027
- Skills: Prompt Engineering, Copilot API Architecture, Secure-Org Policy, Enterprise AI Integration, Workflows
- GitHub Foundations** Issued: Sep 2024 – Expires: Sep 2027
- Skills: Version Control Tools, Git, CI/CD, Repository Management
- ConnectWise Automate — Certified Enterprise Scripting Architect** Issued: Aug 2024
- Skills: RMM, Automation, PowerShell, Windows Server, Enterprise Architecture, ConnectWise Automate